

Project Design Phase Solution Architecture

Date	06/09/2026
Team ID	ServiceNow
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Automated Network Request Management — Solution Architecture Diagram:

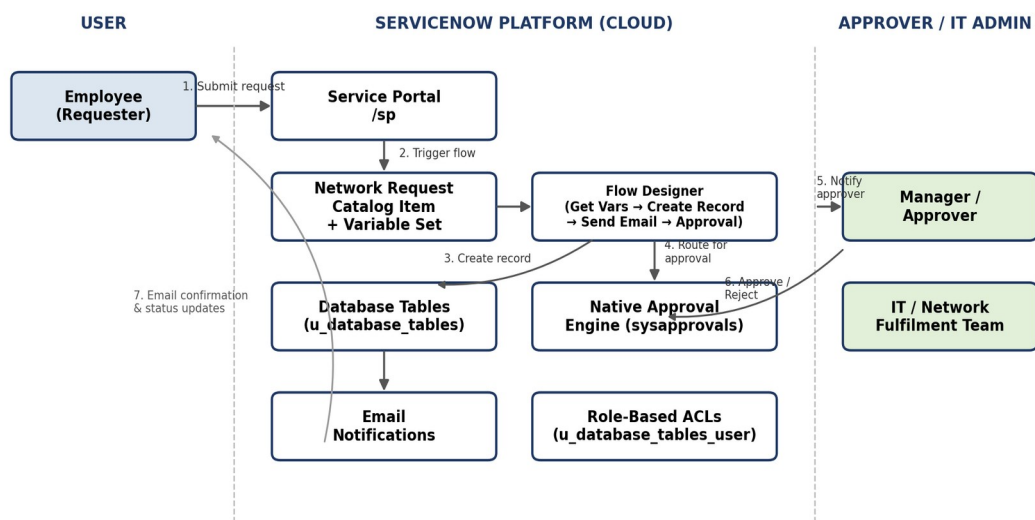


Figure 1: Architecture and data flow of the Automated Network Request Management solution on ServiceNow

The solution follows a straightforward flow: the employee submits a request through the Service Portal's Network Request catalog item, which is bound to a Request Information variable set that auto-populates requester details. On submission, a Flow Designer flow retrieves the catalog variables, creates a record on the custom Database Tables (u_database_tables) table, sends a confirmation email, and routes the request to the requester's manager through ServiceNow's native Approval engine. Once a decision is made, the flow updates the request's Approval Status and notifies the requester by email; approved requests are then actioned by the IT / Network Fulfilment Team. Access to request records is restricted to the u_database_tables_user role through auto-generated ACLs.

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>